

# Variational Quantum Regression

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The start of our Variational Quantum Regression (VQS) is a solution known as an "ansatz". An ansatz is a quantum circuit with parameters that are used to "approximate" a solution.

We start with two data sets  $x, y$ , where  $x$  is independent and  $y$  is dependent. Start by fitting linear data to a classical model:

$$y = mx + b \quad (1)$$

We parameterize the ansatz with parameters  $a$  and  $b$  and feed it into our quantum computer. We get a cost function output from the quantum computer and feed said cost function into the classical computer. This will improve our ansatz, we keep repeating this process until we have a satisfactory ansatz.

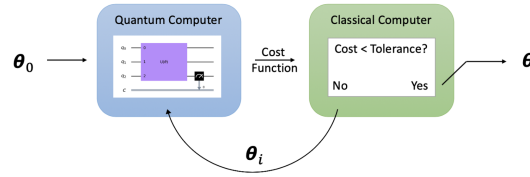


Figure 1: Image imported from Qiskit textbook on github.com

We initialize the qubits into a  $|y\rangle$  state. We can think of  $y$  as a  $N$  length vector.

$$|y\rangle = \frac{1}{C_y} \vec{y} \quad (2)$$

Where  $C_y$  is a normalization factor.

Our ansatz parameterized by  $a$  and  $b$  is described by:

$$|\Phi\rangle = \frac{1}{C_\Phi} (a\vec{x} + b) \quad (3)$$

Again  $C_\Phi$  is a normalization factor.

Now according to github our cost function is:

$$C_P = (1 - \langle y | \Phi \rangle)^2 \quad (4)$$

We next use our quantum computer to calculate the inner product  $\langle x | y \rangle$ . Once we have calculated the inner product we then use the classical computer to update our a and b parameters. We keep repeating this process until we have optimized our cost function.

The process of calculating the inner product of our two data sets involves applying a Hadamard gate to the first qubit to place it in a state of superposition, then find their associated probabilities. Next repeat by applying a Phase gate to the first qubit and measuring its probability as well. When we combine the two probabilities we are finally able to find the true inner product:

$$\langle x | y \rangle \tag{5}$$

We are able to do this process over and over until we reach the desired result. VQR is important in helping us solve regression problems in everyday life. It manipulates quantum circuits to model complex relationships found in data sets. It also is being used by scientist to search for and find a quantum advantage. This could be a breakthrough for high dimensional structures and data sets. We can do this by keeping most of our code the same but updating our cost function to fit higher order polynomials:

$$y = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \tag{6}$$